

SPECIAL REPORT

It's one of the food industry's great success stories. A chemist working on an ulcer medicine during the 1960s casually licks the powder on his finger and finds it sweet. Nearly 20 years later the substance, aspartame, sweetens foods such as breakfast cereal, chewing gum, cocoa, instant iced tea, and whiskey sour mix. As Equal, it's a granulated sugar substitute; as NutraSweet, it sweetens Diet Coke, Diet Pepsi, and Diet Seven-Up.

But some scientists and consumer groups say the additive may cause headaches, dizziness, and depression in adults and brain damage to developing fetuses. Still, the Food and Drug Administration insists that aspartame is safe. And the sweetener's manufacturer, G. D. Searle & Company, touts aspartame as "the most-tested food additive in history."

Aspartame was first approved by the FDA in 1974. But the FDA quickly suspended its approval because new animal studies seemed to indicate that aspartame might cause brain damage.

A 1980 FDA panel of two neurologists and a nutritional biochemist, looking at 15 of Searle's roughly 80 safety studies, said the sweetener would not cause brain damage in consumers. But they could not rule out the possibility that aspartame was carcinogenic and recommended the FDA continue to withhold approval.

In 1981, the new FDA director, Arthur Hayes, who has since left the agency, overruled the panel's decision and approved the additive for dry food and as a granulated sugar substitute. Hayes relied in part on a new study by a Japanese producer of aspartame that showed the additive did not produce tumors. Last summer aspartame was approved for use in soft drinks.

Aspartame is made of aspartic acid and phenylalanine, two of the 20 or so amino acids that are used by the body to manufacture proteins. The two amino acids are present in many foods. But unlike most proteins, which are large and therefore take some time to digest, aspartame is a small molecule and is

ASPARTAME: SOME BITTER WITH THE SWEET?



by William F. Allman

readily broken down, rapidly releasing phenylalanine and aspartic acid into the blood.

Richard Wurtman, an endocrinologist at the Massachusetts Institute of Technology, says that consuming large amounts of aspartame can lead to a sudden rise in phenylalanine, a chemical precursor to neurotransmitters that affect mood and behavior. This increase, says Wurtman, may be linked to depression, sleep problems, loss of vision, headaches, and even seizures. Aspartame also appears to block the production of an appetite-controlling neurotransmitter called serotonin, says Wurtman, which may not be exactly what dieters want. Instead of easing a hunger for sweets, aspartame may cause a craving for more.

The FDA rejected Wurtman's study. "We looked at the data," says Anthony Brunetti, an FDA consumer safety officer on aspartame. "We saw changes in some levels of chemicals. But the ques-

tion is, are they significant changes? We don't think so."

But Wurtman and the Community Nutrition Institute point to more than 1,000 letters from people complaining of neurological disorders after eating food containing aspartame. The FDA says the letters, while anecdotal evidence, are not scientific proof.

"There's nothing wrong with anecdotes from a scientific viewpoint," says the institute's executive director Rod Leonard. "They're simply experience gained from the use of a product and are a valid indicator of problems." The institute is now suing the FDA to reopen the aspartame case.

According to UCLA endocrinologist William Pardridge, a rise in phenylalanine in the bloodstream may cause other problems. Pardridge cites a report by Harvey Levy of Boston's Children's Hospital that as many as 10 percent of the population may have difficulty metabolizing phenylalanine and not know it. While this would not affect most people, pregnant women with such a disorder may put their babies at risk. A high level of phenylalanine in mothers has been shown to cause mental retardation in their babies. It may be possible, says Levy, that a slight increase in phenylalanine would affect the development of the fetus' brain, decreasing the IQ of the infant as much as 15 points.

Brunetti says these people are at no greater risk consuming aspartame than they would be eating foods that contain phenylalanine naturally, such as milk and beef. For the rest of the population, it would be nearly impossible to eat enough aspartame to raise their phenylalanine to harmful levels.

One clue to why they can't get enough aspartame can be found on the label of many soft drinks, though in some cases it is in fine print. Most soft drinks mix NutraSweet with saccharin. Saccharin improves the shelf life of a soda; it's also cheaper than aspartame. Saccharin, of course, has its own checkered history: It causes cancer in laboratory animals. S